GATE-AIII BRANCHES
Engineering Mathematics



SINGLE VARIABLE CALCULUS

Lecture No.- 06



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Recape of previous lecture



Topic How to evaluate the limits?

Topic Concept of continuity

Topic Questions based on continuity

Topics to be covered



- Topic: Concept of continuity — Problem
- Topic: Problems based on continuity
- Topic: Types of discontinuity
- Topic: Maxima and minima
- Topic: Problems based on maxima and minima

Concept of continuity :

Without Lifting my Pen



Mathematically

Condition for continuous function

$$LHL = RHL = f(a)$$

$$LHL = \lim_{h \to 0} f(a-h) \qquad RHL = \lim_{h \to 0} f(a+h)$$

$$\lim_{h \to 0} f(a-h) = \lim_{h \to 0} f(a+h) = f(a)$$

$$\lim_{h \to 0} LHL = \lim_{h \to 0} RHL = f(a)$$

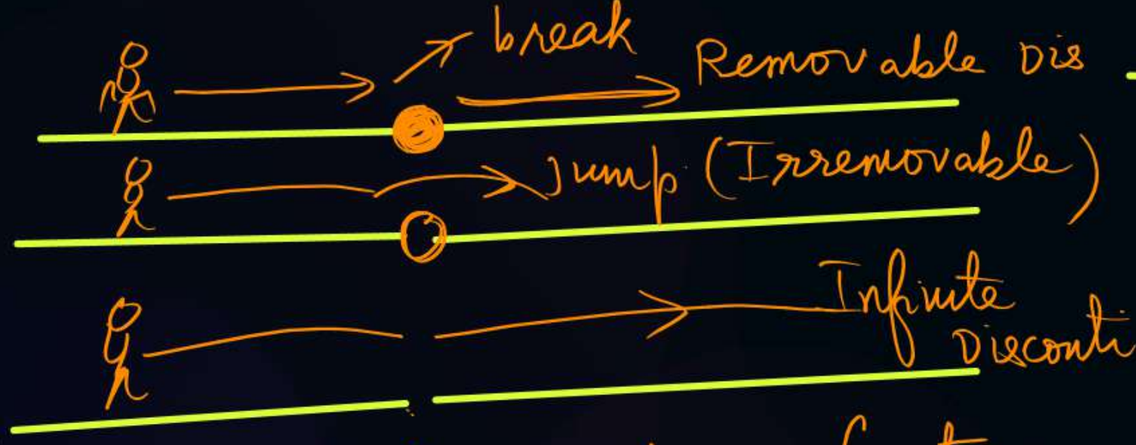
→ condition of continuity

Condition of continuity

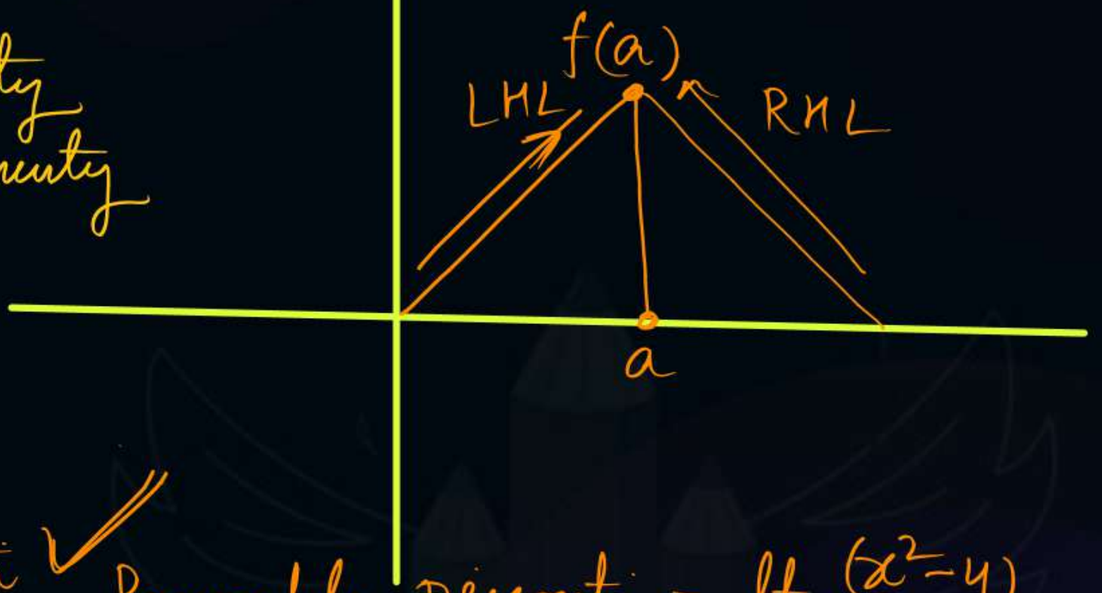


Types of Discontinuity :

1) Removable Discontinuity
2) Irremovable Discontinuity



→ Not removed for Discontinuous function (oscillations)



Removable Disconti $= \lim_{x \to 2} \frac{(x^2-4)}{(x-2)}$

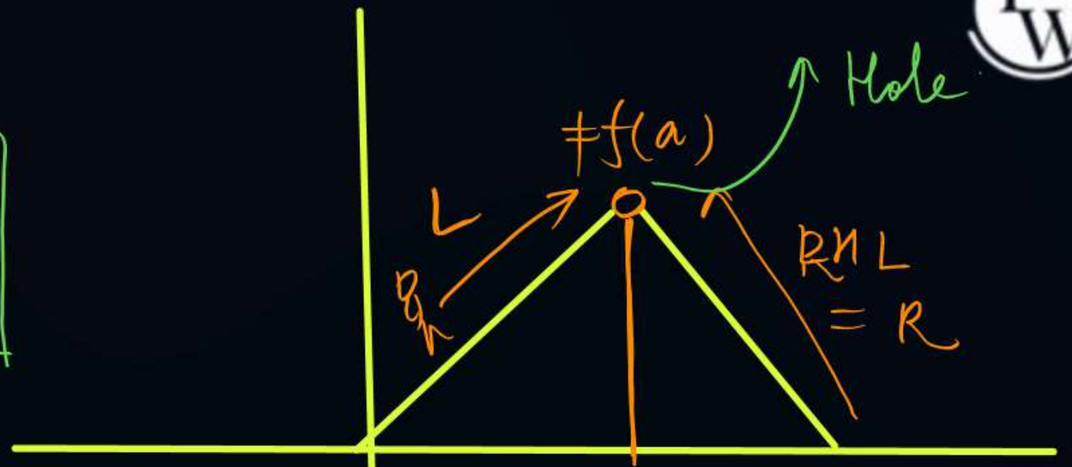
$= \lim_{x \to 2} \frac{(x-2)(x+2)}{(x-2)}$

continuous

Removable Discontinuity :

$$LHL = RHL \neq f(a)$$

$$\lim_{h\to 0} f(a-h) = \lim_{h\to 0} f(a+h) \neq f(a)$$



Jump Discontinuity :

$$LHL \neq RHL$$

$$\text{Jump} = |RHL - LHL|$$

Infinite Discontinuity : (Irremovable discontinuity)

Does Not exist

(Discontinuous)

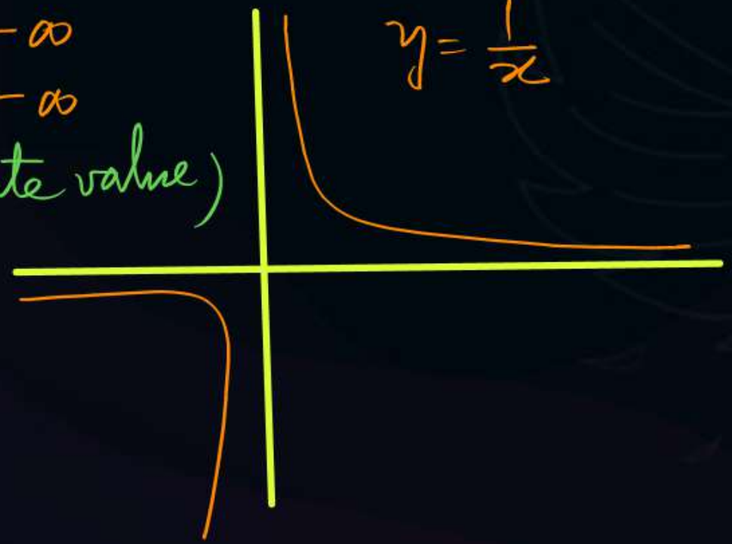
$LHL = -\infty \quad LHL = -\infty$

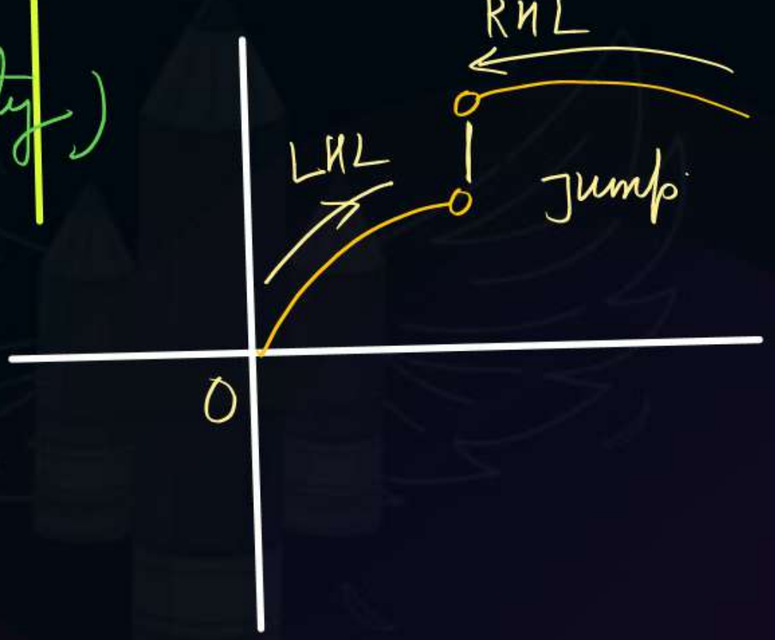
$RHL = +\infty \quad RHL = -\infty$

$RHL = +\infty$

$LHL = +\infty$

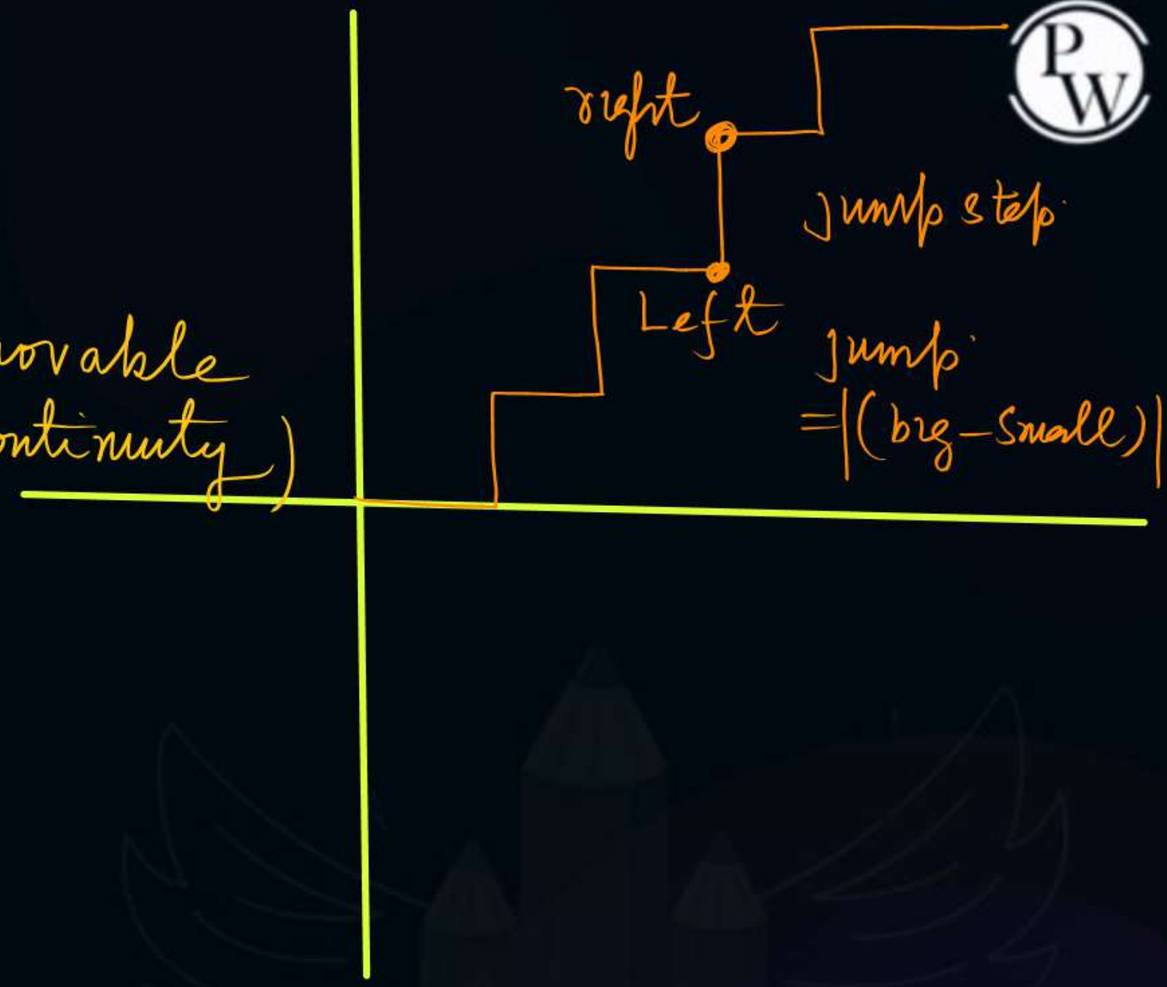
(finite value)





Jump Discontinuity

$LHL \neq RHL$ (Irremovable Discontinuity)

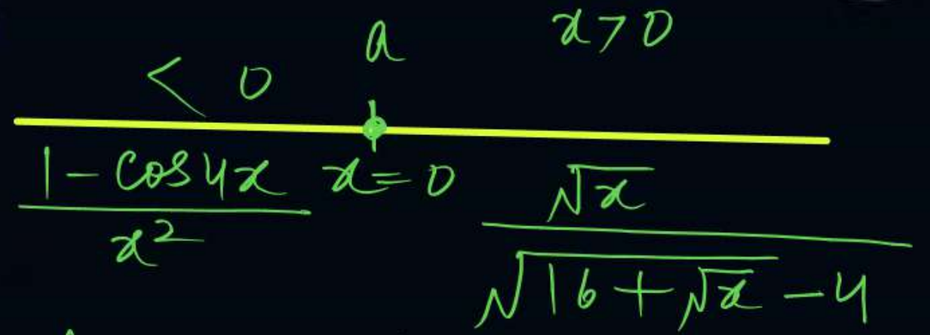




Topic : Single Variable Calculus

#Q. Let $f(x)=\begin{cases} \dfrac{1-\cos 4x}{x^2} & ; \quad x<0 \\ a & ; \quad x=0 \\ \dfrac{\sqrt{x}}{\sqrt{16+\sqrt{x}}-4} & ; \quad x>0 \end{cases}$

The value of a, if possible, so that the function is continuous at $x = 0$ is:



Function is continuous at $x=0$
find the value of 'a'

A 6

B 8

C −6

D None of these

1) Condition for continuity $\quad LHL = RHL = f(0) = a$

2) Calculate the LHL and RHL

$$LHL = \lim_{h\to 0} f(a-h) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0} f(-h)$$

a is Unknown

$(-h)$

$$f(x) = \frac{1-\cos 4x}{x^2}$$

$$= \lim_{h\to 0} \frac{1-\cos 4h}{h^2} = \frac{\to 0}{\to 0} \text{ form}$$

remove the $\frac{\to 0}{\to 0}$

Then apply L-Hospital Rule

$$f(-h) \Rightarrow \frac{1-\cos 4(-h)}{(-h)^2}$$

$$= \frac{1-\cos 4h}{h^2}$$

$$= \lim_{h\to 0} \frac{\frac{d}{dh}(1-\cos 4h)}{\frac{d}{dh}(h^2)}$$

$$= \lim_{h\to 0} \frac{+\sin 4h \cdot 4 \times 2}{2h \times 2} = \lim_{h\to 0} \left(\frac{\sin 4h}{4h}\right) \cdot 8 = 1 \times 8 = 8$$

$$RHL = \lim_{h\to 0} f(a+h) = \lim_{h\to 0} f(0+h) = \lim_{h\to 0} f(h)$$

$$= \lim_{h\to 0} \frac{\sqrt{h}}{\sqrt{16+\sqrt{h}}-4} \left(\frac{\to 0}{\to 0}\right) \quad f(x) = \frac{\sqrt{x}}{\sqrt{16+\sqrt{x}}-4}$$

$$= \lim_{h\to 0} \frac{\sqrt{h}}{\sqrt{16+\sqrt{h}}-4} \times \frac{\sqrt{16+\sqrt{h}}+4}{\sqrt{16+\sqrt{h}}+4}$$

$$= 8$$

$$LHL = RHL = f(a)$$

$$8 = 8 = a$$

$$\boxed{a = 8} \quad \text{Ans.}$$



Topic : Single Variable Calculus

5 – 10 Min
Rahul Sir PW

#Q. The values a and b so that the function:

$$f(x)=\begin{cases} x+a\sqrt{2}\sin x & ; \quad 0\le x<\dfrac{\pi}{4} \\ 2x\cot x+b & ; \quad \dfrac{\pi}{4}\le x\le\dfrac{\pi}{2} \\ a\cos 2x-b\sin x & ; \quad \dfrac{\pi}{2}<x\le\pi \end{cases}$$

is continuous $x\in[0,\pi]$ is:

A $a=\dfrac{\pi}{6},b=\dfrac{-\pi}{12}$

B $a=\dfrac{\pi}{3},b=\dfrac{-\pi}{12}$

C $a=\dfrac{\pi}{6},b=\dfrac{\pi}{12}$

D None of these

at $a = \frac{\pi}{4}$

$$LHL = \lim_{h\to 0} f(a-h) = \lim_{h\to 0} f\left(\frac{\pi}{4}-h\right) \quad x \to \frac{\pi}{4}-h$$

$$= \lim_{h\to 0} \left(\frac{\pi}{4}-h\right) + a\sqrt{2}\sin\left(\frac{\pi}{4}-h\right)$$

$$= \frac{\pi}{4} + a\sqrt{2}\sin\frac{\pi}{4} = \frac{\pi}{4} + a\sqrt{2}\times\frac{1}{\sqrt{2}}$$

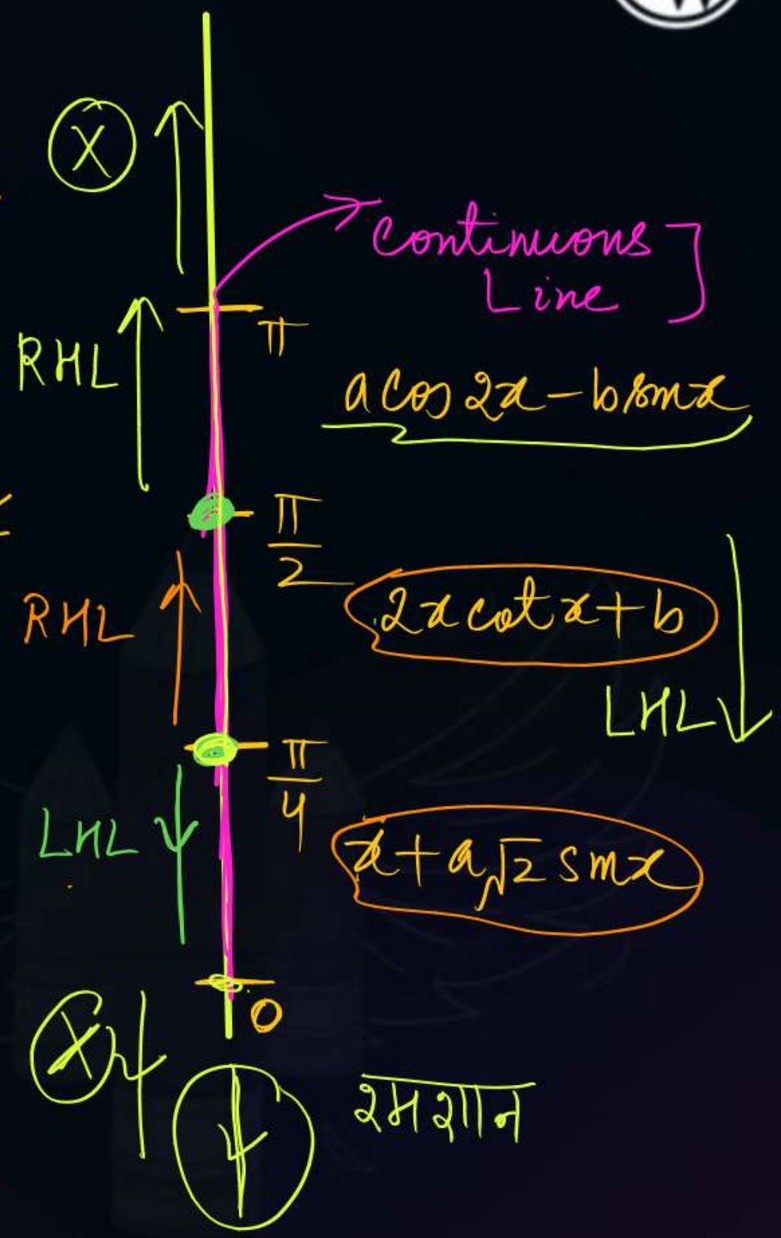
$$\boxed{LHL = \frac{\pi}{4} + a}$$

$x \to \frac{\pi}{4}+h$

$$RHL = \lim_{h\to 0} f(a+h) = \lim_{h\to 0} f\left(\frac{\pi}{4}+h\right)$$

$$\lim_{h\to 0} 2\left(\frac{\pi}{4}+h\right)\cot\left(\frac{\pi}{4}+h\right) + b$$

$$\underline{RHL} = \frac{\pi}{2}\cot\left(\frac{\pi}{4}\right) + b = \frac{\pi}{2} + b$$



According to condition of continuity

$$LHL = RHL$$

$$\frac{\pi}{4} + a = \frac{\pi}{2} + b$$

$$a - b = \frac{\pi}{2} - \frac{\pi}{4} = \frac{4\pi - 2\pi}{8}$$

$$\boxed{a - b = \frac{\pi}{4}} \quad -(1)$$

at point $\frac{\pi}{2}$ $\quad a + 2b = 0 \quad -(2)$

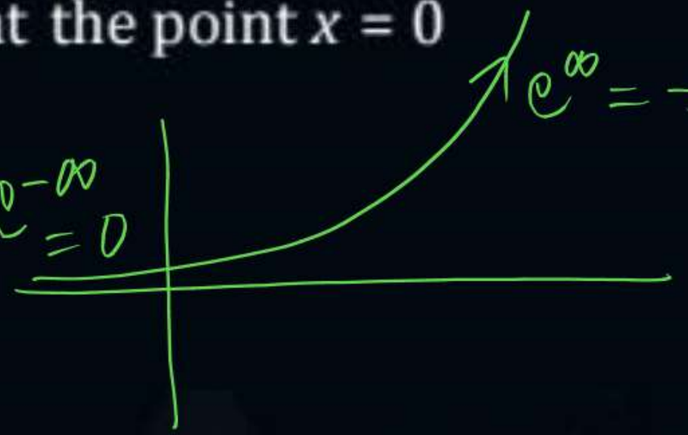
$$\boxed{a = \frac{\pi}{6} \quad b = -\frac{\pi}{12}}$$ Ans

Topic : Single Variable Calculus

Discontinuity

#Q. Discuss the continuity of $f(x)=\begin{cases}\dfrac{e^{1/x}-1}{e^{1/x}+1} & ; \quad x\neq 0 \\ 0 & ; \quad x=0\end{cases}$ at the point $x=0$

$$f(x)=\begin{cases}\dfrac{e^{\frac{1}{x}}-1}{e^{1/x}+1} & x\neq 0 \\ 0 & x=0\end{cases}$$



$$\text{Left Hand limit} = \lim_{h\to 0} f(a-h) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0} f(-h)$$

$$= \lim_{h\to 0} \frac{e^{-\frac{1}{h}}-1}{e^{-1/h}+1} = -1$$

$$RHL = \lim_{h\to 0} f(a+h) = \lim_{h\to 0} f(0+h) = \lim_{h\to 0} f(h)$$

$$= \lim_{h\to 0} \frac{e^{\frac{1}{h}} - 1}{e^{1/h} + 1}$$

Multiply $e^{-\frac{1}{h}}$
Divide via $e^{1/h}$

$$= \lim_{h\to 0} \frac{1 - e^{-1/h}}{1 + e^{-1/h}} \quad (e^{-1/h} \to 0)$$

$$= 1$$

$LHL \neq RHL$ this a jump discontinuity



Topic : Single Variable Calculus

5 – 7 min

#Q. Let $f(x)=\begin{cases} (1+|\sin x|)^{\frac{a}{|\sin x|}} & ; \quad \frac{-\pi}{6}<x<0 \\ b & ; \quad x=0 \\ e^{\left(\frac{\tan 8x}{\tan 3x}\right)} & ; \quad 0<x<\frac{\pi}{6} \end{cases}$

Left

right

The value a and b such that $f(x)$ is continuous at $x = 0$ is:

A $a=8, b=e^{8}$

B $a=\frac{8}{3}, b=e^{-8}$

C $a=\frac{8}{3}, b=e^{8/3}$

D None of these

$$RHL = \lim_{h\to 0} f(a+h) = \lim_{h\to 0} f(0+h) = \lim_{h\to 0} f(h)$$

$$= \lim_{h\to 0} e^{\frac{\tan 8h}{\tan 3h}}$$

$$= \lim_{h\to 0} e^{\left(\frac{\tan 8h}{8h}\right)\frac{3h}{\tan 3h}\left(\frac{8h}{3h}\right)}$$

$$\boxed{b = e^{8/3}}$$

$$LHL = \lim_{h\to 0} f(a-h) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0} f(-h)$$

$$\lim_{h\to 0}\left[1+|-\sin h|\right]^{\frac{a}{|-\sin h|}} \quad \lim_{h\to 0}\left[1+|\sin(-h)|\right]^{\frac{a}{|\sin(-h)|}}$$

$$= \lim_{h\to 0}\left[1+\sin h\right]^{\frac{a}{\sin h}} = e^a$$

$$e^a = e^{8/3} = b$$

$$a = \frac{8}{3}$$

$$b = e^{8/3}$$



Topic : Single Variable Calculus

#Q. If $f(x)=\dfrac{1}{1+e^{1/x}}, x\neq 0$, discuss the continuity of $f(x)$ at $x=0$.

$$f(x)=\frac{1}{1+e^{1/x}} \quad x\neq 0 \quad \text{at } x=0$$

$$RHL=\lim_{h\to 0} f(0+h)=\lim_{h\to 0} f(h)=\lim_{h\to 0}\left\{\frac{1}{1+e^{1/h}}\right\}=0$$

$$LHL=\lim_{h\to 0} f(0-h)=\lim_{h\to 0} f(-h)=\lim_{h\to 0}\left[\frac{1}{1+e^{-1/h}}\right]=1$$

$RHL=0$, $LHL=1$ } Jump discontinuity

Jump $=|0-1|=1$

$\frac{1}{1+e^{\infty}}=\frac{1}{1+\vec{\infty}}=\frac{1}{\infty}=\vec{0}$

$\frac{1}{1+e^{-\infty}}=\frac{1}{1+\vec{0}}=1$



2 mins Summary

- Topic One
- Topic Two
- Topic Three
- Topic Four
- Topic Five

Condition for continuity LHL = RHL = f(a)

Removable Discontinuity LHL = RHL ≠ f(a) Indeterminate form

Limit

Jump " = LHL ≠ RHL

Infinite Discontinuity LHL, RHL] both $+\infty$ $-\infty$]

MEAN value Theorem

- → Differentiability
- → Rolles Theorem
- → Lagrange Theorem
- → Max/min with one variable

THANK - YOU